

A sharper distortion bound for the minimal faithful representation of $C_2 \times C_2$

Candidate partial result for arXiv:2506.04425

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Status. Candidate partial result, likely valid. This packet improves the upper bound for one distinguished representation; it does not compute the group contortion.

1 Source question and contribution

Blum-Smith, Derksen, Mixon, Qaddura, and Vose define the Euclidean contortion of a finite group by

$$\Upsilon(G) = \sup_{V \in \text{Rep}(G)} c_2(V/G),$$

and ask in Section 4.2 for the values of $\Upsilon(C_4)$ and $\Upsilon(C_2 \times C_2)$ [1]. The question appears on page 43:

for the wallpaper groups with orbifold signature 333, 3*3, 442, or 632, the quotient metric is unique up to a scale factor, and yet Theorem 26 fails to establish the Euclidean distortion.

In light of Theorem 7 it would be interesting to compute the Euclidean contortion $\Upsilon(G)$ for groups beyond those handled in Lemma 8. For example, what are the values of $\Upsilon(C_4)$ and $\Upsilon(C_2 \times C_2)$? More ambitiously, if one computed the contortions of all finite simple groups, Theorem 9 would produce bounds on the contortions of all finite groups in terms of their simple subquotients. Should we expect such contortion bounds to be relatively tight? Or do there exist many groups with low contortion whose simple subquotients have large contortion? On the asymptotic side, what is the infimum of α for which $\Upsilon(G) = O(|G|^\alpha)$ for all finite groups G ? See the end of Section 2.2 for further discussion.

Here $c_2(X)$ is the least distortion of a bilipschitz embedding of the metric space X into a Hilbert space. The source proves $\sqrt{2} \leq \Upsilon(C_2 \times C_2) \leq 2$. We study the natural minimal faithful representation

$$G := \{\text{diag}(1, 1, 1), \text{diag}(1, -1, -1), \text{diag}(-1, 1, -1), \text{diag}(-1, -1, 1)\} \curvearrowright \mathbb{R}^3.$$

This is the multiplicity-free direct sum of the three nontrivial real characters of $C_2 \times C_2$.

2 Proof intuition

An orbit is determined by the three absolute coordinates and, away from the coordinate planes, the parity of the signs. Thus $\mathbb{R}^3/G$ is two copies of the positive orthant glued along their boundary. On the same sheet the quotient distance is Euclidean. Between sheets, the best group element leaves exactly one sign mismatch, in the coordinate minimizing $a_i b_i$.

This suggests embedding the glued orthants by keeping $a = (|x_1|, |x_2|, |x_3|)$ and adding one signed coordinate equal to $q(a) := \min_i a_i$, the Euclidean distance from a to the boundary. The resulting same-sheet and cross-sheet squared distances are explicit. A two-variable scalar inequality bounds their ratio between $2 - \sqrt{2}$ and 2. The quotient of these two constants is $2 + \sqrt{2}$.

The original contortion problem remains open because a general $C_2 \times C_2$ -representation may contain arbitrary multiplicities of all three nontrivial characters.

3 Metric model

Put $C = \mathbb{R}_+^3$ and $q(a) = \min_i a_i$. For $a \in C$ with all coordinates positive, let $\varepsilon \in \{\pm 1\}$ record the sign of the product of the coordinates of a representative. If $q(a) = 0$, this sign is immaterial because the orbit contains representatives of both parities. Hence the orbit space is identified with

$$X = (C \times \{\pm 1\}) / ((a, +1) \sim (a, -1) \text{ when } q(a) = 0).$$

Lemma 1 (Exact quotient metric). *For $a, b \in C$, set*

$$A = \|a - b\|_2^2, \quad m = \min_{1 \leq i \leq 3} a_i b_i.$$

Then

$$d_X((a, \varepsilon), (b, \delta))^2 = \begin{cases} A, & \varepsilon = \delta, \\ A + 4m, & \varepsilon = -\delta. \end{cases}$$

Proof. For equal parity, an even sign change aligns all coordinate signs, giving distance $\|a - b\|_2$, and coordinatewise absolute values show that no smaller distance is possible. For opposite parity, any alignment has an odd number of sign mismatches. If the mismatched coordinates form S , its squared distance is

$$A + 4 \sum_{i \in S} a_i b_i.$$

All summands are nonnegative, so the minimum over odd S is obtained by a singleton at an index attaining m . $\square$

4 The embedding and its bound

Define

$$F : X \longrightarrow \mathbb{R}^4, \quad F(a, \varepsilon) = (a, \varepsilon q(a)).$$

It is well-defined on the glued boundary since $q(a) = 0$ there.

Lemma 2 (The scalar inequality). *For $a, b \in C$, with A, m as in Lemma 1, and $x = q(a), y = q(b)$, one has*

$$A + (1 + \sqrt{2})(x + y)^2 \geq 4\sqrt{2}m.$$

Proof. The assertion is immediate if $m = 0$. By homogeneity, assume $m = 1$. Then $a_i b_i \geq 1$ for each i , and $xy \leq 1$. Define

$$f(t) = \begin{cases} (t^{-1} - t)^2, & 0 < t < 1, \\ 0, & t \geq 1. \end{cases}$$

Choose an index r with $a_r = x$. If $x < 1$, then $b_r \geq 1/x$, so $(a_r - b_r)^2 \geq f(x)$. Similarly, an index where b attains y contributes at least $f(y)$. If the two indices coincide, $xy = 1$, and at most one of $f(x), f(y)$ is nonzero. Consequently,

$$A \geq f(x) + f(y). \quad (1)$$

Write $\lambda = 1 + \sqrt{2}$. First suppose $x, y < 1$ and put $s = xy \leq 1$. Using $x^{-2} + y^{-2} \geq 2/s$, $x^2 + y^2 \geq 2s$, and $(x + y)^2 \geq 4s$, (1) gives

$$\begin{aligned} A + \lambda(x + y)^2 &\geq \frac{2}{s} - 4 + (2 + 4\lambda)s \\ &\geq 4\sqrt{2}. \end{aligned}$$

Indeed, the last one-variable expression is minimized at $s = (1 + \sqrt{2})^{-1}$.

If, say, $x \geq 1$, then $y \leq 1$. Decreasing x to 1 only decreases the remaining lower bound, and

$$f(y) = \left(\frac{1}{y} - y\right)^2 \geq 4(1 - y)^2.$$

Thus

$$A + \lambda(x + y)^2 \geq 4(1 - y)^2 + \lambda(1 + y)^2 \geq \frac{16\lambda}{4 + \lambda} \geq 4\sqrt{2}.$$

The case $y \geq 1$ is symmetric; if both are at least 1, then $x = y = 1$ and the conclusion is immediate. $\square$

Theorem 1 (Improved bound for the minimal faithful representation). *For the above action of $G \cong C_2 \times C_2$ on $\mathbb{R}^3$,*

$$c_2(\mathbb{R}^3/G) \leq \sqrt{2 + \sqrt{2}}.$$

Together with the lower bound in Theorem 24 of [1], this gives

$$\sqrt{2} \leq c_2(\mathbb{R}^3/G) \leq \sqrt{2 + \sqrt{2}} < 2.$$

Proof. For points on the same sheet, Lemma 1 and the fact that $q = \min_i a_i$ is 1-Lipschitz give

$$d_X^2 = A \leq \|F(a, \varepsilon) - F(b, \varepsilon)\|^2 = A + (x - y)^2 \leq 2A.$$

For points on opposite sheets, put

$$D^2 = \|F(a, \varepsilon) - F(b, -\varepsilon)\|^2 = A + (x + y)^2, \quad d_X^2 = A + 4m.$$

Since $A \geq (x - y)^2$ and $m \geq xy$,

$$2d_X^2 - D^2 = A + 8m - (x + y)^2 \geq 4xy \geq 0.$$

Set $c = 2 - \sqrt{2} = \sqrt{2}(\sqrt{2} - 1)$. Lemma 2 yields

$$\begin{aligned} D^2 - cd_X^2 &= (\sqrt{2} - 1)A + (x + y)^2 - 4(2 - \sqrt{2})m \\ &= (\sqrt{2} - 1)(A + (1 + \sqrt{2})(x + y)^2 - 4\sqrt{2}m) \geq 0. \end{aligned}$$

Thus every squared distance ratio lies in $[2 - \sqrt{2}, 2]$. Therefore

$$\kappa(F)^2 \leq \frac{2}{2 - \sqrt{2}} = 2 + \sqrt{2},$$

as claimed. $\square$

5 Optimality inside the signed-boundary family

For $t > 0$, consider $F_t(a, \varepsilon) = (a, t\varepsilon q(a))$. Same-sheet pairs that differ only in their smallest coordinate have squared distance ratio exactly $1 + t^2$. On the other hand, take opposite-sheet points

$$a = (1, L, L), \quad b = (L, 1, L), \quad L = 1 + \sqrt{2}.$$

Here $A = 4$, $m = L$, and $q(a) = q(b) = 1$, so their squared distance ratio is

$$\frac{4 + 4t^2}{4 + 4L} = \frac{1 + t^2}{2 + \sqrt{2}}.$$

It follows that $\kappa(F_t)^2 \geq 2 + \sqrt{2}$ for every $t > 0$. Hence the choice $t = 1$ in Theorem 1 is optimal throughout this natural one-parameter family.

6 Verification and limitations

The script `code/verify_bound.py` samples random pairs in both sheets, checks the exact quotient formula by enumeration of the four group elements, checks the proved ratio interval, and evaluates the extremal family above. It is a regression check, not part of the proof.

The theorem does not determine $c_2(\mathbb{R}^3/G)$ exactly and does not upper-bound $\Upsilon(C_2 \times C_2)$ below 2, since contortion takes the supremum over all finite-dimensional representations. The exact phrases “Euclidean contortion”, “Upsilon(C4)”, and “C2 times C2”, together with the paper title and arXiv id, were searched on 27 August 2026. No later resolution or matching bound was found. This is a bounded novelty check only.

Human review recommendation. Check Lemma 2, especially the case where the minima of a and b occur at the same coordinate, and independently verify the cross-sheet metric formula. If those pass, retain as a partial result; do not relabel it as a solution of the contortion question.

References

- [1] B. Blum-Smith, H. Derksen, D. G. Mixon, Y. Qaddura, and B. Vose, *Estimating the Euclidean distortion of an orbit space*, arXiv:2506.04425v1 (2025).